

● HARD

● PROBLEM-SOLVING AND DATA ANALYSIS

Problem-Solving and Data Analysis

10

8 questions · ~12 min

Data set A consists of 10 positive integers less than 60. The list shown gives 9 of the integers from data set A.

43, 45, 44, 43, 38, 39, 40, 46, 40

The mean of these 9 integers is 42. If the mean of data set A is an integer that is greater than 42, what is the value of the largest integer from data set A?

STUDENT-PRODUCED RESPONSE

.	/.	/.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q3

GRID-IN

PROBLEM-SOLVING AND DATA ANALYSIS

The number a is 110% greater than the number b . The number b is 90% less than 47. What is the value of a ?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q4

PROBLEM-SOLVING AND DATA ANALYSIS

The result of increasing the quantity x by 400% is 60. What is the value of x ?

- A. 12
- B. 15
- C. 240
- D. 340

Q5

PROBLEM-SOLVING AND DATA ANALYSIS

140 is $p\%$ greater than 10. What is the value of p ?

- A. 1,400
- B. 1,300
- C. 140
- D. 130

Year	Subscriptions sold
2012	5,600
2013	5,880

The manager of an online news service received the report above on the number of subscriptions sold by the service. The manager estimated that the percent increase from 2012 to 2013 would be double the percent increase from 2013 to 2014. How many subscriptions did the manager expect would be sold in 2014?

- A. 6,020
- B. 6,027
- C. 6,440
- D. 6,468

Q7

PROBLEM-SOLVING AND DATA ANALYSIS

In 2008, Zinah earned 14% more than in 2007, and in 2009 Zinah earned 4% more than in 2008. If Zinah earned y times as much in 2009 as in 2007, what is the value of y ?

- A. 0.5600
- B. 1.0056
- C. 1.1800
- D. 1.1856

Q8

GRID-IN

PROBLEM-SOLVING AND DATA ANALYSIS

The number w is 110% greater than the number z . The number z is 55% less than 50. What is the value of w ?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.